

PRACTICAL-5

QUESTION 1 - INNER JOIN

```
mysql> SELECT d.name AS Department,
->          p.name AS Head_Physician
-> FROM department d
-> JOIN physician p
-> ON d.head = p.employeeid;
```

Department	Head_Physician
General Medicine	Percival Cox
Surgery	John Wen
Psychiatry	Molly Clock

3 rows in set (0.00 sec)

QUESTION 2 - WHERE CLAUSE

```
mysql> SELECT *
-> FROM physician
-> WHERE position='Surgical Attending Physician';
```

employeeid	name	position	ssn
3	Christopher Turk	Surgical Attending Physician	333333333
6	Todd Quinlan	Surgical Attending Physician	666666666
7	John Wen	Surgical Attending Physician	777777777

3 rows in set (0.01 sec)

QUESTION 3 - WHERE + LIKE

```
mysql> SELECT *
-> FROM physician
-> WHERE name LIKE 'John%';
```

employeeid	name	position	ssn
1	John Dorian	Staff Internist	111111111
7	John Wen	Surgical Attending Physician	777777777

2 rows in set (0.00 sec)

QUESTION 4 - COUNT()

```
mysql>
mysql> SELECT COUNT(*) AS Total_Physicians
-> FROM physician;
+-----+
| Total_Physicians |
+-----+
|                9 |
+-----+
1 row in set (0.01 sec)
```

QUESTION 5 - COUNT DISTINCT

```
mysql> SELECT COUNT(DISTINCT position)
-> FROM physician;
+-----+
| COUNT(DISTINCT position) |
+-----+
|                        7 |
+-----+
1 row in set (0.00 sec)
```

QUESTION 6 - GROUP BY

```
mysql> SELECT position,
-> COUNT(*) AS Total_Employees
-> FROM physician
-> GROUP BY position;
+-----+-----+
| position                | Total_Employees |
+-----+-----+
| Staff Internist         | 1               |
| Attending Physician     | 1               |
| Surgical Attending Physician | 3               |
| Senior Attending Physician | 1               |
| Head Chief of Medicine  | 1               |
| MD Resident             | 1               |
| Attending Psychiatrist  | 1               |
+-----+-----+
7 rows in set (0.00 sec)
```

QUESTION 7 - GROUP BY + HAVING

```
mysql> SELECT position,
->      COUNT(*) AS Total_Employees
-> FROM physician
-> GROUP BY position
-> HAVING COUNT(*) > 1;
```

position	Total_Employees
Surgical Attending Physician	3

1 row in set (0.01 sec)

QUESTION 8 - ORDER BY

```
mysql> SELECT *
-> FROM physician
-> ORDER BY name ASC;
```

employeeid	name	position	ssn
5	Bob Kelso	Head Chief of Medicine	555555555
3	Christopher Turk	Surgical Attending Physician	333333333
2	Elliot Reid	Attending Physician	222222222
1	John Dorian	Staff Internist	111111111
7	John Wen	Surgical Attending Physician	777777777
8	Keith Dudemeister	MD Resident	888888888
9	Molly Clock	Attending Psychiatrist	999999999
4	Percival Cox	Senior Attending Physician	444444444
6	Todd Quinlan	Surgical Attending Physician	666666666

9 rows in set (0.01 sec)

QUESTION 9 - ORDER BY DESC

```
mysql> SELECT *
-> FROM physician
-> ORDER BY employeeid DESC;
```

employeeid	name	position	ssn
9	Molly Clock	Attending Psychiatrist	999999999
8	Keith Dudemeister	MD Resident	888888888
7	John Wen	Surgical Attending Physician	777777777
6	Todd Quinlan	Surgical Attending Physician	666666666
5	Bob Kelso	Head Chief of Medicine	555555555
4	Percival Cox	Senior Attending Physician	444444444
3	Christopher Turk	Surgical Attending Physician	333333333
2	Elliot Reid	Attending Physician	222222222
1	John Dorian	Staff Internist	111111111

9 rows in set (0.00 sec)

QUESTION 10 - WHERE + IN

```
mysql> SELECT *
-> FROM physician
-> WHERE employeeid IN
-> (
->     SELECT head
->     FROM department
-> );
```

employeeid	name	position	ssn
4	Percival Cox	Senior Attending Physician	444444444
7	John Wen	Surgical Attending Physician	777777777
9	Molly Clock	Attending Psychiatrist	999999999

3 rows in set (0.00 sec)

QUESTION 11 - SUBQUERY

```
mysql> SELECT *
-> FROM physician
-> WHERE employeeid NOT IN
-> (
->     SELECT head
->     FROM department
-> );
```

employeeid	name	position	ssn
1	John Dorian	Staff Internist	111111111
2	Elliot Reid	Attending Physician	222222222
3	Christopher Turk	Surgical Attending Physician	333333333
5	Bob Kelso	Head Chief of Medicine	555555555
6	Todd Quinlan	Surgical Attending Physician	666666666
8	Keith Dudemeister	MD Resident	888888888

6 rows in set (0.01 sec)

QUESTION 12 - GROUP BY + ORDER BY

```
mysql> SELECT position,
->     COUNT(*) AS Total_Employees
-> FROM physician
-> GROUP BY position
-> ORDER BY Total_Employees DESC;
```

position	Total_Employees
Surgical Attending Physician	3
Staff Internist	1
Attending Physician	1
Senior Attending Physician	1
Head Chief of Medicine	1
MD Resident	1
Attending Psychiatrist	1

7 rows in set (0.00 sec)

QUESTION 13 - WHERE + LIKE

```
mysql> SELECT *
-> FROM physician
-> WHERE position LIKE '%Attending%';
```

employeeid	name	position	ssn
2	Elliot Reid	Attending Physician	222222222
3	Christopher Turk	Surgical Attending Physician	333333333
4	Percival Cox	Senior Attending Physician	444444444
6	Todd Quinlan	Surgical Attending Physician	666666666
7	John Wen	Surgical Attending Physician	777777777
9	Molly Clock	Attending Psychiatrist	999999999

6 rows in set (0.00 sec)

QUESTION 14 - JOIN + WHERE

```
mysql> SELECT p.name
-> FROM physician p
-> JOIN department d
-> ON p.employeeid=d.head
-> WHERE d.name='Surgery';
```

name
John Wen

1 row in set (0.00 sec)

QUESTION 15 - JOIN + ORDER BY

```
mysql> SELECT d.name,
-> p.name
-> FROM department d
-> JOIN physician p
-> ON d.head=p.employeeid
-> ORDER BY d.name;
```

name	name
General Medicine	Percival Cox
Psychiatry	Molly Clock
Surgery	John Wen

3 rows in set (0.00 sec)

QUESTION 16 - WHERE + NOT LIKE

```
mysql> SELECT *
-> FROM physician
-> WHERE position NOT LIKE '%Surgical%';
```

employeeid	name	position	ssn
1	John Dorian	Staff Internist	111111111
2	Elliot Reid	Attending Physician	222222222
4	Percival Cox	Senior Attending Physician	444444444
5	Bob Kelso	Head Chief of Medicine	555555555
8	Keith Dudemeister	MD Resident	888888888
9	Molly Clock	Attending Psychiatrist	999999999

6 rows in set (0.00 sec)

QUESTION 17 - COUNT + HAVING

```
mysql> SELECT position,
-> COUNT(*) AS Total
-> FROM physician
-> GROUP BY position
-> HAVING COUNT(*) >= 2;
```

position	Total
Surgical Attending Physician	3

1 row in set (0.00 sec)

QUESTION 18 - EXISTS

```
mysql> SELECT *
-> FROM department d
-> WHERE EXISTS
-> (
-> SELECT 1
-> FROM physician p
-> WHERE p.employeeid=d.head
-> );
```

departmentid	name	head
1	General Medicine	4
2	Surgery	7
3	Psychiatry	9

3 rows in set (0.00 sec)

QUESTION 19 - MAX()

```
mysql> SELECT MAX(employeeid)
-> FROM physician;
+-----+
| MAX(employeeid) |
+-----+
|                9 |
+-----+
1 row in set (0.00 sec)
```

QUESTION 20 - MIN()

```
mysql> SELECT MIN(employeeid)
-> FROM physician;
+-----+
| MIN(employeeid) |
+-----+
|                1 |
+-----+
1 row in set (0.00 sec)
```